



LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Study Plan for Degree Project

Degree Project: MASK11, Mathematical Statistics
Student: Nancy Truong
Advisor: Johan Lindström
Examiner: Ted Kronvall
Time: 15 January 2026 – 31 May 2026
Preliminary examination date: 10 June 2026

Content:

To study the foraging habits of bumblebees and other pollinating insects, artificial hives are used to track insect movement and collect harvested pollen. An important ecological question is which plant species the insects have visited. This can be studied by collecting pollen grains and classifying these using machine learning methods.

In a previous bachelor thesis by Erik Jurdell, a convolutional neural network (CNN) was used to classify pollen species from microscopic images. In addition to image data, size features were incorporated into the model to improve classification recall. While the results were strong for many species, some species proved substantially harder to classify than others, probably due to class imbalances and intrinsic class difficulty.

My preliminary research questions are:

- How can hierarchical clustering and difficulty-aware learning objectives be used to improve the classification of pollen species?
- Can existing image-based pollen classification methods be extended and improved by explicitly accounting for class imbalances and class difficulty, with particular focus on species that were difficult to identify in previous work?

Literature:

Tian, Junjiao; Mithun, Niluthpol; Seymour, Zach; Chiu, Han-Pang; Kira, Zsolt, Striking the Right Balance: Recall Loss for Semantic Segmentation (June 28, 2021). Available at arXiv: <https://arxiv.org/abs/2106.14917>

Lin, Tsung-Yi; Goyal, Priya; Girshick, Ross; He, Kaiming; Dollár, Piotr, Focal Loss for Dense Object Detection (August 7, 2017). Available at arXiv: <https://arxiv.org/abs/1708.02002>

Please, turn over!

Ranglani, Hardev, Residual Aware Stacking: A Novel Approach for Improved Machine Learning Model Performance (January 18, 2025). Available at SSRN: <https://ssrn.com/abstract=5160281> or <http://dx.doi.org/10.2139/ssrn.5160281>

Time table:

- 15 January – 10 March: Theory and background study. Literature review on image-based pollen classification, convolutional neural networks, feature extraction, class imbalance, difficulty-aware loss functions (e.g. focal and recall-based loss), and hierarchical statistical models. Familiarization with Erik's thesis, dataset structure, and experimental setup. Reimplementation of the baseline CNN model with size information.
- 11 March – 30 April: Implementation and experimentation. Development of proposed extensions, including hierarchical clustering of CNN-extracted features, difficulty-aware loss functions, and hierarchical classification models. Model training, validation, and comparative experiments. Concurrent writing of the methodology, experiments, and preliminary results sections.
- 1 May – 31 May: Finalization and presentation. Completion of experiments and final model comparisons. Analysis and interpretation of results, including performance on hard-to-classify species. Final writing, revision, and proofreading of the report in collaboration with the advisor. Submission of the thesis and preparation for the oral seminar presentation.

This study plan has been signed in two copies by student and the advisor. The signed study plan should be emailed by the student to the relevant Programme Administrator; i.e. Malin Ejlerskov (for Mathematics) or Susann Nordqvist (for Mathematical Statistics).

Lund, 17 February 2026

Student

Advisor